DE.AUTO

CHAPTER 11

AERO & LIGHTNESS

THE WORLD OF CAR MODIFICATIONS MADE SIMPLE

Cars explained in plain, friendly English

MEMBER EXCLUSIVE · DE.AUTO ·

POWERED BY DE.AUTO

DE.AUTO — Because knowledge is power, and cars are passion.

CAR MODIFICATIONS GUIDE
MUST-KNOW TERMS

Car modifications, explained

Chapter Eleven: Aerodynamics and Weight Reduction

“SPEED IS BUILT ON LESS AIR AND LESS WEIGHT.”

— DE.AUTO

Important notice

This guide is **entertainment only**. It is not educational or scientific content, and it is not a how-to manual. DE.AUTO is not an accredited scientific, engineering or technical authority. Everything here is simplified for fun and general interest, and it can differ from one car to another.

You must **never** try any modification, drive at high speed, or do any kind of stunt or dangerous driving on public roads. Safety and respecting traffic laws are everyone's own responsibility.

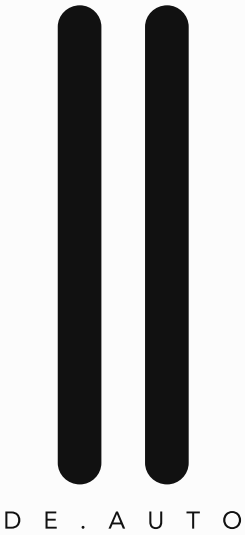
The information here is **general and approximate**, and much of it describes only the typical case. Laws, regulations and specifications differ by country and model, so always check the owner's manual and the official authorities.

Vehicle modifications are subject to local laws. Any unauthorized modification can lead to legal liability, fines or having your vehicle impounded.

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For any decision about your car, go to certified professionals and the manufacturer's manual, and only try something if it complies with official regulations and legal requirements. 🚀



DE.AUTO is more than a name. It's an identity that brings the passion for cars together with knowledge and progress.

At DE.AUTO we make fresh content for car lovers, mixing fun and know-how from the world of modifications and high performance, in a simple style that's easy to follow. Our goal is to help you understand your own car better and make smarter calls on modifications and maintenance.

We walk you through the key terms and parts of the modification world, so you can chat with the enthusiasts and follow every detail without feeling like an outsider.

Whether you're a beginner or a pro, you'll find terms, basics and fun facts here that help you get into cars in an organized, confident way.

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READY?

A huge wing on the boot... and it might still do nothing? 🤨 And one kilo off a wheel beats several kilos out of the boot?

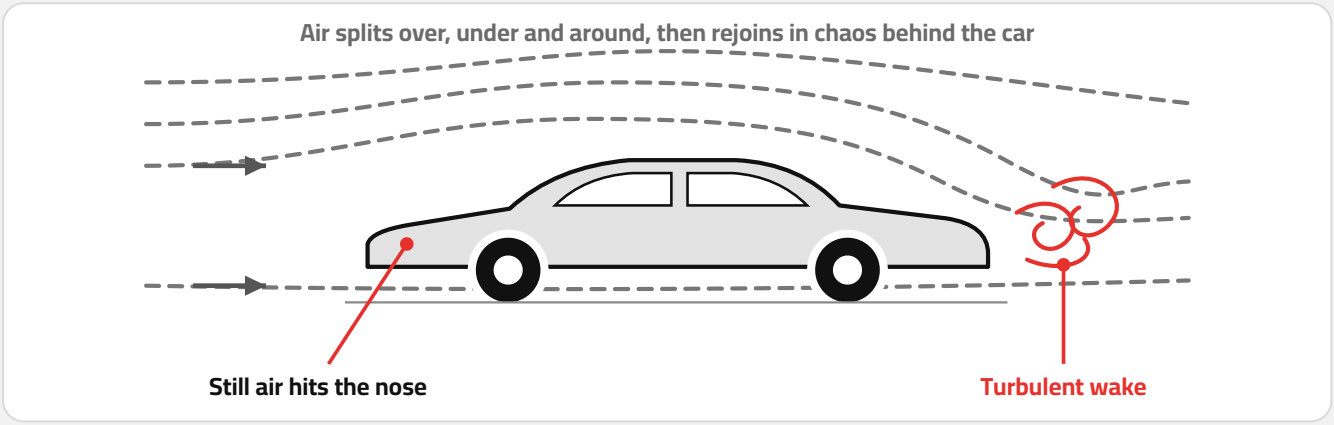
By the end of this chapter you'll know:

- ✓ What is air to a car, and why does it push back harder and harder as speed rises?
- ✓ The difference between Drag, Lift and Downforce, in drawings
- ✓ Splitter, canard, side skirts, diffuser: what each does and where it goes
- ✓ The real difference between a **spoiler** and a **wing**
- ✓ What actually works on the road and what is only for show (honestly)
- ✓ Power-to-weight with a worked example, and why rotating weight costs more than static weight
- ✓ Lightweight materials (aluminium, carbon, magnesium), their price, and the safety limits you never cross



Air... heavier than you think 🌬️

You don't feel the air while you walk, but your car feels it from the first moment. Air is a **fluid** like water, only lighter, and whenever you move through it you have to "push it aside". That pushing aside is called **aerodynamic drag**.



Simplified illustration of the flow; reality is far more complex

The drag equation, made simple

The force that resists your car is set by four things. Don't fear the symbols, they're all logical:

Air density (ρ)
The denser the air, the more resistance. Hot summer air in Riyadh is a little lighter than air on a cold day.

Frontal area (A)
The size of the "face" that hits the air: roughly width × height. A tall, wide car pushes more air.

Drag coefficient (Cd)
A number that says how "slippery" the shape is. The lower, the smoother.

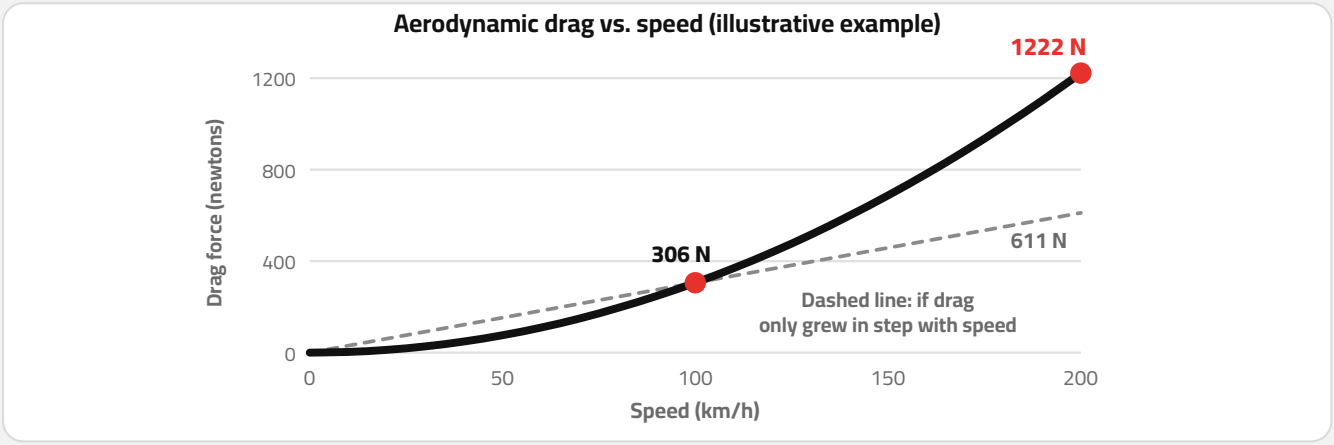
Speed (v)
Here's the big secret: it enters the equation **squared**, and we'll see its effect on the next page.

0.25–0.40 The approximate range of the drag coefficient Cd in passenger cars. A flat plate facing the airflow head-on goes above 1. Shape makes all the difference.

🔍 **Small detail, big effect:** Manufacturers work on seemingly simple things like mirror shape, door edges and underbody covers, just to lower the Cd by a few points. And the wheels and cooling openings account for a large part of the drag.

The secret of the square: why speed makes everything costlier

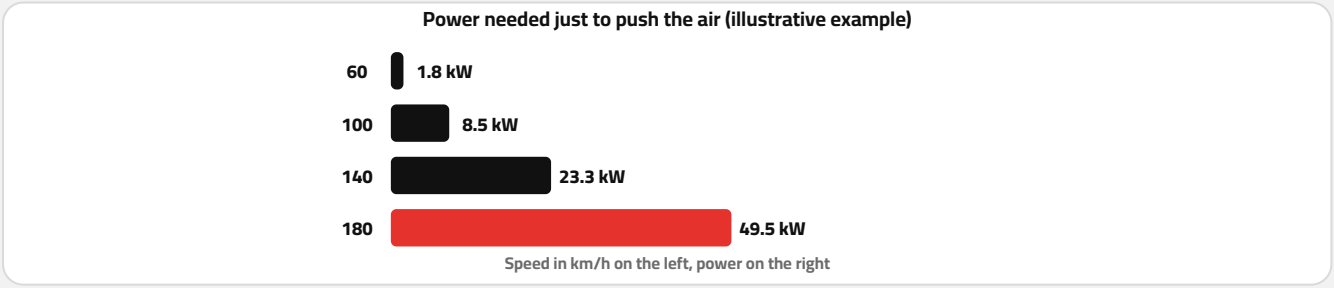
One rule explains half of this chapter: drag force grows with the **square** of the speed. Double your speed and the drag becomes **four times** as big. Even harsher, the power the engine needs to push through that resistance becomes **eight times**.



Illustrative example: a hypothetical car with Cd 0.30 and a frontal area of 2.2 m². The numbers are theoretical physics, not an invitation to speed. Obey road limits.

×8

The power the car needs to fight the air if you double your speed. In our example: roughly 8.5 kW at 100 km/h, and roughly 68 kW at 200 km/h (approximate calculations for a hypothetical car).



Air only, without tire friction and drivetrain losses

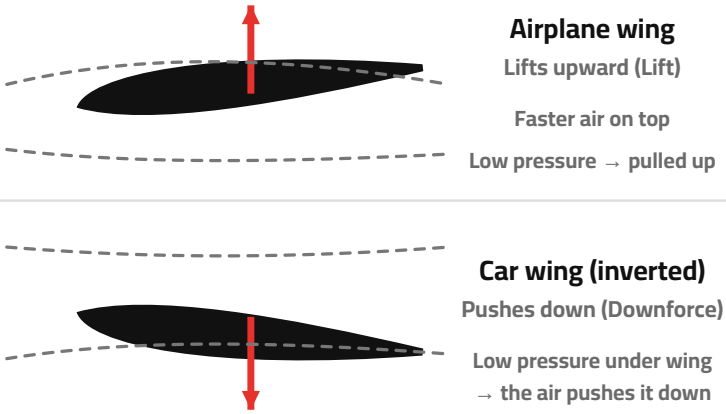
🧳 **An example from your life:** This is why track cars and record cars work on every millimetre of shape. And it's also why driving with a roof box or an empty roof rack on the highway really does raise fuel consumption, so remove it if you don't need it.

🎯 **Challenge 1** **If you double your speed, what roughly happens to aerodynamic drag?**

A) It doubles (×2) B) It becomes ×4 C) It doesn't change

Lift or downforce? Downforce without weight

The air passing over the car's body doesn't only create drag. It also gives it a vertical force: upward is called **lift**, and downward is called a **downward force** or **downforce**.



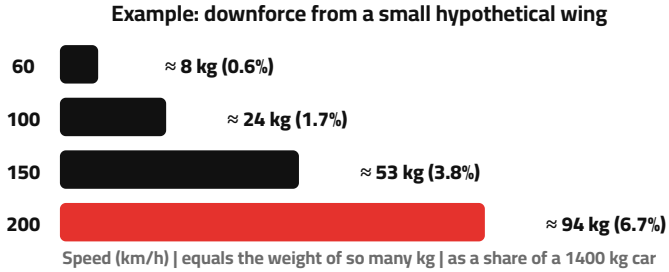
Wing cross-section (simplified). The same idea, inverted

Why do we want downforce?

A tire grips better as the load on it increases. Downforce **adds load without adding mass**, so it doesn't make the car heavier, and it goes away when you slow down.

The cost

Any wing that presses down usually generates **extra drag** (Induced Drag). You gain grip and pay with top speed and fuel.



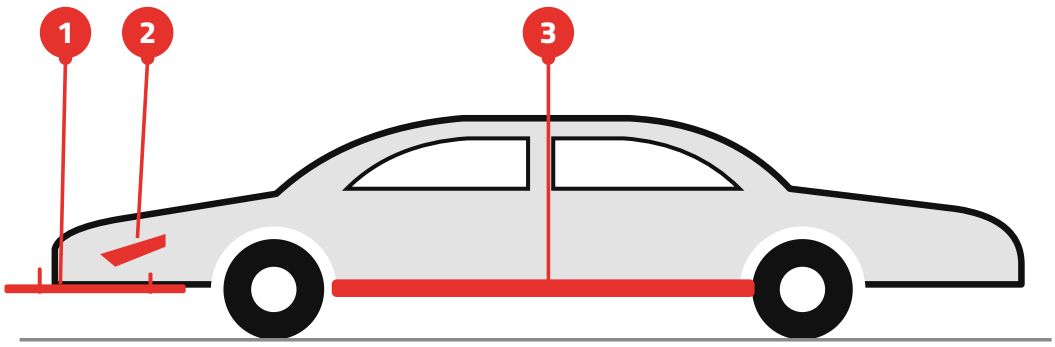
Illustrative example: the numbers change completely with design, angle and installation

⚖️ **To be honest:** A tire doesn't gain grip in a straight line as load rises. The first bit of load adds clear grip, but each increase after it gives less than the one before (Load Sensitivity). So downforce is useful, but it's not magic. The details are in Chapters **8** and **10**.

🔥 **The same square law:** Since it also follows the **square** of speed, downforce at 60 km/h is hardly felt at all. Even in our example, at 100 km/h it equals roughly 2% of the car's weight.

Front and sides: where does the air go?

As soon as the air hits the nose you have two options: it goes over, or it goes under. What goes under the car is usually **harmful**: turbulent, and it lifts the front. So the first job is at the air dam:



Nose and sides: three things that steer air away from under the car

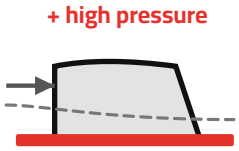
Simplified drawing; real shapes differ

1. Splitter and lip

The splitter is a flat board that sticks out below the bumper and splits the air. The lip is a shorter version attached to the bumper.

2. Canard

Small fins on the side of the bumper that create vortices to manage the air around the wheel. On road cars their effect is usually small.



The splitter board: higher pressure above it than below, which presses the nose down

3. Side skirts

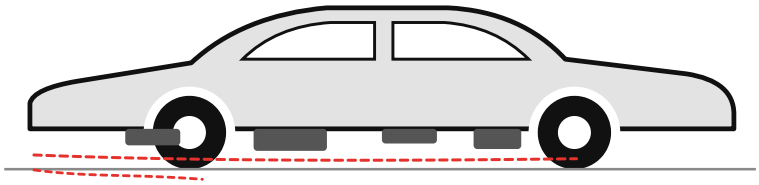
Panels along the side of the car between the wheels reduce air getting under the car from the side. Also called Side Skirts, and sometimes just “skirts”.

❌ **Closer to the ground = better?** No. A low splitter scrapes on bumps, kerbs and ramps, and the lip breaks and bumpers get damaged. And many can’t take the air pressure if the mounting is weak.

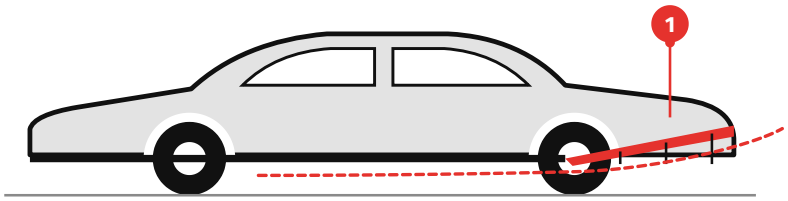
⚖️ **Balance:** If you fit a front splitter without a wing or balance at the rear, the front gets pressed down and the rear stays light. The balance between front and rear matters more than the size of any one part. We’ll come back to it.

The underbody and the diffuser: the hidden force

In some race cars designed for ground effect, a big part of the downforce comes from **underneath the car**, not only from the big wing. There's a whole world of air you never see.



Rough underbody: air hits the parts and trips



Flat floor + diffuser: smooth, expanding flow

Simplified comparison. A real underbody includes exhaust, shafts and cooling

Flat underbody: panels that cover hanging parts and let air pass without tripping and at speed. They cut drag and organise the pressure under the car.

Diffuser: a channel that widens toward the rear. The air under the floor moves faster so its pressure drops and pulls the car down, and the diffuser lets that air slow down gradually and return to atmospheric pressure at the exit, which keeps the flow under the car stable. This is the basis of what's called **ground effect**.

✕ MYTH

"I stuck a plastic diffuser under the rear bumper and the car sticks to the road now."

✓ FACT

A diffuser only works with **a flat floor ahead of it** and a channel designed at a specific angle. If the underside is exposed in front of it, the part is most likely decoration. Some just give a sporty look.

🎯 Challenge 2 **What does a diffuser do?**

A) Raises exhaust temperature

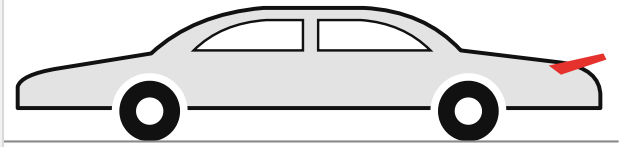
B) Helps the air under the car slow down smoothly and keeps the pressure under it low

C) Reduces the car's weight

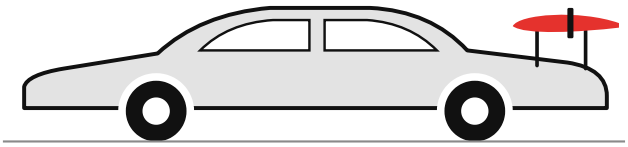
Spoiler or wing? It's not just a name

People mix the two up, even in shops. But the idea is completely different:

Spoiler



Wing



	Spoiler	Wing
The idea	It spoils **the airflow** and reduces lift at the rear	An inverted aerofoil section that produces **downforce** directly
Position	Attached to the boot surface or edge	Raised on stands in clean air
Drag	May reduce drag slightly if carefully designed	Usually increases drag
Speed needed	Its effect is small and modest	Its big effect shows at high speed

Ducktail

A small lip on the edge of the boot (ducktail), lighter and simpler than a big wing.

Signs of a "looks-only" wing

A very thin piece, close to the boot, with a fixed angle and no data, usually on stands too weak to take the pressure.

❌ **A wing at the rear only.** The rear gets pressed down and the front goes relatively light, and the car's character changes at high speed (the front can lose its ability to grip). That's why track people balance front and rear together.

❌ **Fitting a heavy wing on a thin boot lid.** The lid warps and water gets in, and rear visibility may suffer. Some wings stick out past the body in breach of regulations. Check local rules and tell your insurer.

🎯 Challenge 3 **What's the key difference between a wing and a spoiler?**

A) A wing produces downforce directly

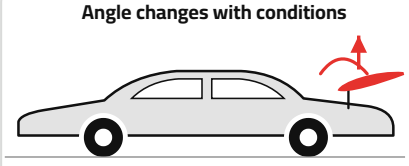
B) A spoiler is always bigger

C) No difference, two names for the same thing

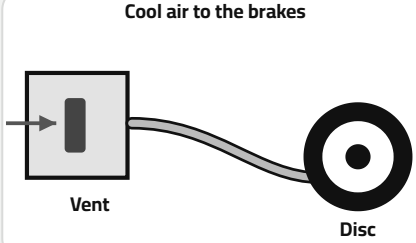
Smart details: fins, flaps and ducts



Small fins on the roof



Moving wing / flap



Vortex generators

Small parts that create vortices to keep the air attached to the surface for longer. Found on the roof of some cars.

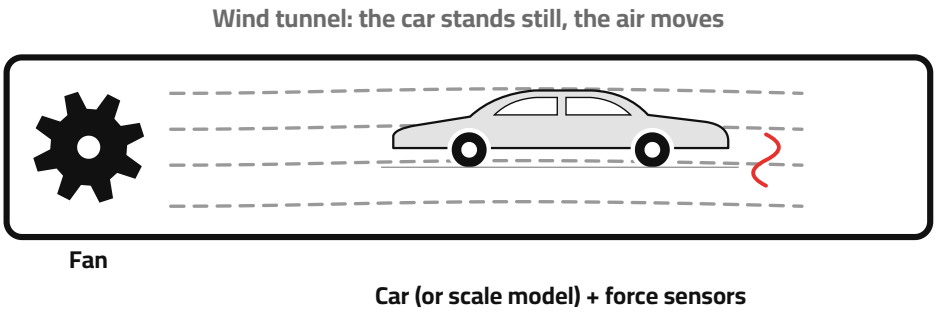
Active aero

Parts that move as needed: open for more cooling, close to cut drag, or raise a wing on the track. Racing cars use it in specific places.

Cooling ducts

Openings that bring air to the radiator and discs. Cooling needs air, and every bit of air let in adds a little drag, so it's a balancing act.

How do they test all this? 🔬



Simplified drawing; real tunnels are more complex, with moving belts

Wind tunnel

A room with a giant fan that blows air over a car or scale model with instruments that measure forces.

Computer simulation (CFD)

Software that calculates the airflow around the car digitally. Saves time and cost, and is compared against real testing.

Honestly: real or just looks?

Most road cars don't travel at speeds where aero makes much difference, and most parts sold as "aero" are **cosmetic**. And that's not a flaw if you know you're buying them for the look. The flaw is paying for a performance promise.

Part	Works for real when…	Looks only when…
Splitter / lip	Often helps at track speeds, with proper balance and mounting	On the road: small effect, and a higher risk of scraping
Canard	On cars designed around it and tunnel-tested	Commercial fins without balance: looks
Side skirts	Reduce air leaking underneath, a small improvement	Cheap and far from the ground: looks
Spoiler / ducktail	Modestly useful on some body shapes	Very small effect at road speeds
Wing	Useful at high speeds on a track	An ordinary wing at legal speeds: usually extra drag and looks
Diffuser	With a flat floor and precise design	A plastic piece under the bumper: decoration
Fake vents	—	Vents that don't actually let air in: decoration

1 : 4

If you compare a car's downforce at 100 km/h with the same car at 200 km/h, the second gives roughly four times as much. So a part that "works" on the track may have almost no effect on your daily route.

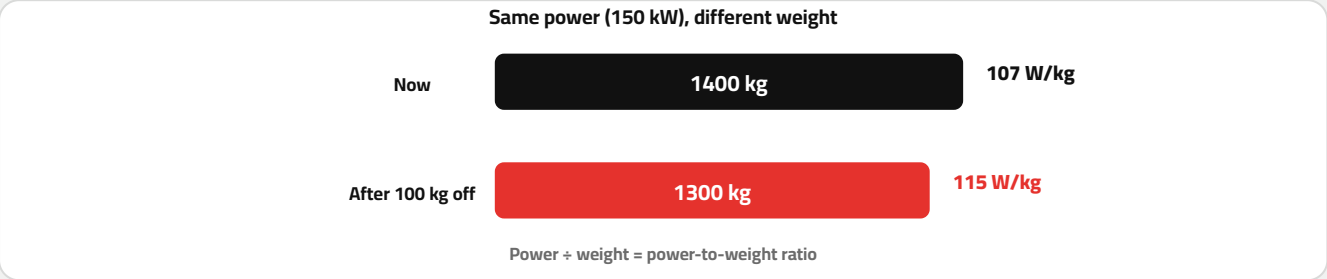
5 questions before you buy any aero part

1. Is there real test data (wind tunnel or measurement), or only pictures and talk?
2. Does the part go with the rest of the car (front and rear)?
3. Is the mounting strong enough to take the load without damaging the body?
4. Does it cover cooling openings or reduce visibility?
5. Is it legal where you live? Does your insurer know?

> 🤝 **DE.AUTO tells you:** There's no performance promise here: the effect changes with the car, the speed and the installation. If you like the look alone, that's a perfectly fine choice, but call things by their names.

Weight: power-to-weight ratio

Every kilo in the car needs force to move it, force to stop it, and force to turn it. That's why makers measure the **power-to-weight ratio**: how many watts (or horsepower) per kilo.



Illustrative worked example. The car's weight here includes the driver and fuel

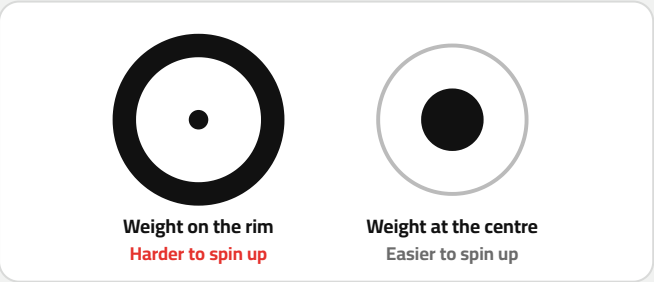
Power **150 kW** (≈ 200 hp), and weight **1400 kg** with the driver: the ratio is roughly 107 W/kg.

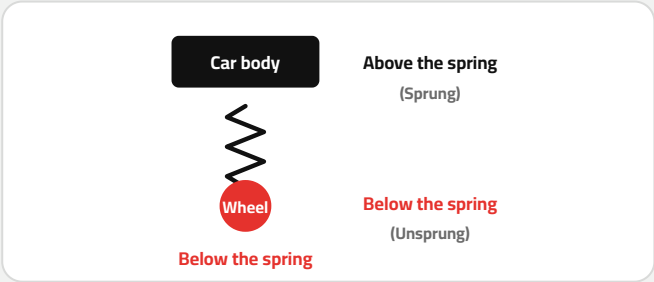
You drop **100 kg**: weight becomes 1300. The ratio is roughly 115 W/kg, an increase of about **7.7%** on paper.

To reach the same ratio without cutting weight, you'd need to add about **11.5 kW** of power (≈ 15 hp). And this comparison is theoretical only.

+7.7% The improvement in the ratio just from taking 100 kg off a 1400 kg car. Your own weight (the driver) counts too!

Rotating and unsprung weight: costlier than the rest ⚙️





Rotating weight

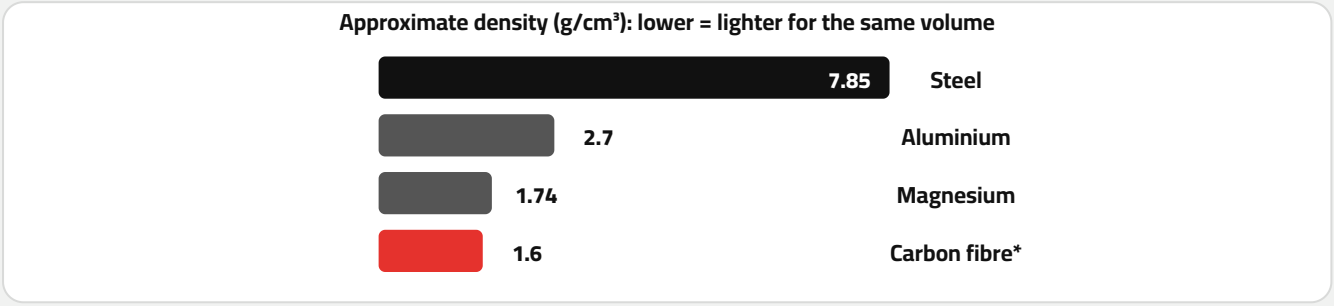
Any part that spins (wheels, discs, gearbox) needs energy to move it forward **and** energy to spin it. That's why one kilo at the rim (wheel/tire) costs more than one static kilo.

Unsprung weight

Parts below the spring (tire, wheel, brakes, and part of the suspension arms) force the suspension to control them directly. Making them lighter lets the wheel follow potholes faster and improves grip. Explained in **Chapter 8**.

Lightweight materials: nothing is free

Cutting weight doesn't mean removing parts, it means swapping steel for a lighter, stronger material depending on use. And every material has a price:



* A composite (fibres + resin), approximate value, varies by type

Material	Advantage	Price and drawback
Aluminium	Relatively light and well-known, resists rust	More expensive than steel; metal fatigue and repairs can be harder
Carbon fibre	Very light and very strong with the right design	Expensive. Internal damage may not show. Hard to repair. Needs UV protection
Magnesium	One of the lightest structural metals	Corrodes, needs surface protection and special handling, and it's expensive

5× Roughly the ratio of steel's density to carbon fibre's as a material (depending on the composite). But lightness doesn't mean cutting the same shape: the part is redesigned to carry the same load.

❌ **A "look" carbon part over a steel part.** Some parts look like carbon but are a thin surface layer (veneer), and their weight may not drop at all. Ask: is the part full carbon or just an outer skin?

❌ **Lighter, cheaper wheels with no certification or testing.** A wheel carries large loads. Cheap ones of unknown origin can crack. The wheel is one of the most important parts for safety. Details are in **Chapter 10**.

🎯 Challenge 4 **Which kind of weight costs more per kilo?**

A) Things in the boot | B) Wheels and rims | C) They're all the same

Stripping out, balance, and safety limits

Stripping the cabin: the price of comfort

Some track people take out interior items. Ask yourself what you lose before you get excited:

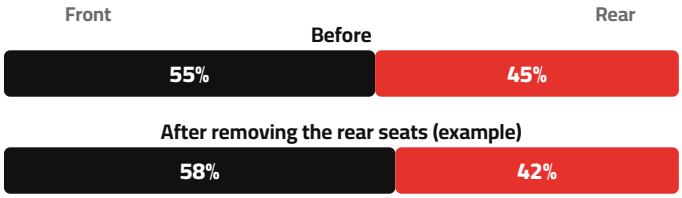
Rear seats and carpet
They save weight, but noise and drone explode, and some seats are part of the safety and registration system.

Sound insulation
It has weight, but removing it makes the cabin loud and tiring on road trips.

Spare tire
Saves weight, but if you get a flat on the road you have no solution. And some cars have no spare in the first place.

Balance after the change ⚖️

Anything you remove or move changes how weight is split front to rear and left to right, the weight distribution. Take the seats out of the back and the front gets relatively heavier, and the car's character changes. Some people move the battery to the rear to balance it.



The numbers are illustrative only

🛑 **Red line: don't remove anything from the safety systems.** Airbags, seat belts, the seat frame and its mounting, the side protection inside the doors, ABS and ESC sensors. These aren't "extra weight", they're your life. And any change that breaks the registration or alters the structure may breach regulations and affect your **insurance**.

Fuel: Less weight usually helps more in the city (stop-start) than on the highway, and the effect is normally small. Aero with higher drag hurts the highway. Choosing one thing doesn't mean losing the other.

Ballast: In racing, class rules set a minimum weight, and anyone under it adds ballast in a specific low position. The idea: fairness between competitors.

Myths, mistakes and the garage glossary

✕ MYTH	✓ FACT
"A big wing = more speed."	A wing usually increases grip at the cost of higher drag. Top speed usually doesn't rise, and may even fall. And at road speeds it doesn't make much difference.
✕ MYTH	✓ FACT
"The lighter the car, the better."	Lightness helps, but if you take it so far that you lose safety, comfort or balance, it works against you. And a heavy, steady car can beat a light, unbalanced one.

❌ We rely on **adverts and photos** instead of tests. For any part promising extra horsepower or km/h with no source, stay away.

❌ We buy a part that **looks like a race car's** and fit it on a daily car, then scrape on ramps and burn more petrol.

Garage talk	Proper name	What it means
Spoyler	Spoiler	Spoils the airflow and reduces lift; a wing is something else
Janah ("wing")	Wing	An inverted section that produces downforce
Donforis	Downforce	The air's pressing force, downward
Difyuzar	Diffuser	A rear channel that helps the air under the car exit smoothly and keeps the pressure low
Spleeter / shifa ("lip")	Splitter / Lip	A board under the front bumper
Sakrata	Side Skirt	A side board between the two wheels
Sahb il-hawa ("air pull")	Drag	Air's resistance to the car's motion
Wazn reesh ("feather weight")	Lightweighting	Cutting weight with materials or by removing items
Wazn il-ajala ("wheel weight")	Unsprung Mass	The weight below the spring (tire, wheel and brakes)

🎯 Challenge 5 **Which change do you never touch, safety-wise?**

A) Carpets B) Airbags C) An extra bonnet cover

Wrap-up 📌

Now you know the car fights the air and its own weight at the same time: drag grows with the square of speed, downforce gives grip without mass, not every wing is useful, and every kilo has a different effect. And most important: safety and the law come before any lightness. 🏁

🎯 Challenge answers:

1) About ×4 (and the power ×8).

2) A wing produces downforce directly.

3) It helps keep the pressure low under the car.

4) Wheels and rims (rotating and unsprung weight).

5) Airbags.

➡️ In **Chapter 12: Car Tuning and ECU**: the car is metal, and the software is its brain. What is the computer that decides everything, and what does “tuning” mean?